

 EXPERIMENTAL FEATURE

Cross-Chat Setup Guide

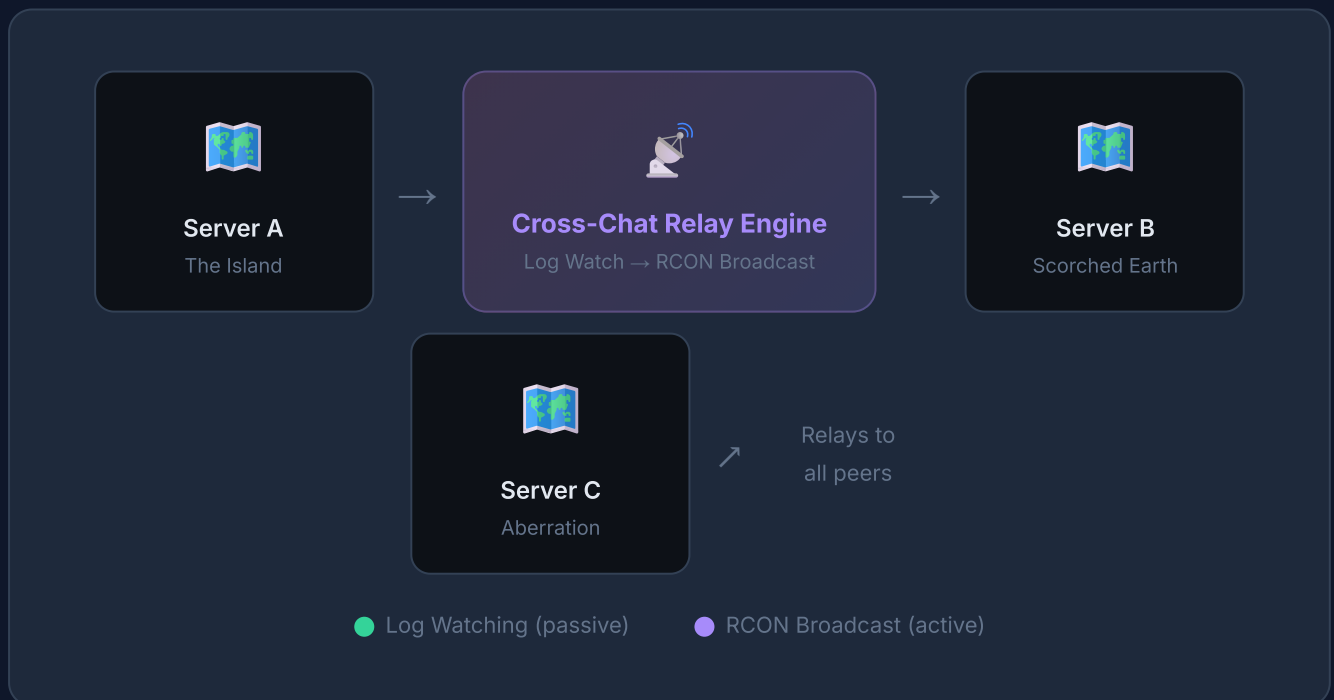
ARK Server Manager 2.0 — Cluster-Wide Chat Relay

Version 2.2.7+ | Last Updated: July 2026

WHAT IS CROSS-CHAT?

Cross-Chat allows players on different maps within the same cluster to see each other's global chat messages in real time. A player on **The Island** can chat seamlessly with a player on **Scorched Earth** — no mods required.

How It Works



The service uses a **Log-driven architecture**:

- 1 Passive Listening** — A `LogWatcher` monitors each `ShooterGame.log` for new chat lines.
- 2 Message Extraction** — A regex parser identifies player chat messages (ignoring system events like saves and join/leave).
- 3 Server-Prefixed Relay** — The message is reformatted as `[ServerName] Player: message` and broadcast to all **RCON**.

 TIP

No mods, no plugins, no binary patches. It runs entirely through existing log files and the RCON protocol. Zero performance impact on game servers.



Prerequisites

Before enabling cross-chat, every server in your cluster **must** meet these requirements:

REQUIREMENT	WHERE TO SET	WHY
RCON Enabled	Server Settings → <code>Enable RCON</code>	Cross-chat uses RCON to broadcast messages
RCON Port	Server Settings → <code>RCON Port</code>	Each server needs a unique RCON port
Admin Password	Server Settings → <code>Admin Password</code>	RCON authentication requires it
In a Cluster	Cluster Manager → Add servers	Servers must be in the same cluster
Servers Running	Start all servers	Log watchers & RCON need active servers

 IMPORTANT

Each server in the cluster **must have a unique RCON port**. If two servers share the same RCON port on the same machine, cross-chat (and RCON in general) will fail.



Step-by-Step Setup

1

Create a Cluster

1. Navigate to **Cluster Manager** from the sidebar.
2. Click the "+ Create Cluster" button (pink button, top-right).
3. Enter a **Cluster Name** (e.g., `My PvE Cluster`).
4. **Browse** and select a shared **Cluster Directory** — this is the folder ARK uses for cross-server character/dino transfers.
Example: `D:\ARKServers\ClusterData`
5. Select the **servers** you want to include from your existing server list.
6. Click **Create**.

2

Verify RCON Settings on Every Server

For **each server** in the cluster, go to Server Settings and confirm:

✓ **RCON Enabled** is ON

✓ **RCON Port** is set **unique per server** e.g., Server A: `27020`, Server B: `27021`, Server C: `27022`

✓ **Admin Password** is set (not empty)

Save any changes.

⚠ WARNING

If RCON is disabled or the admin password is blank on any server, cross-chat **will not work** for that server.

3

Start All Servers

1. Start servers **individually** or use the cluster's **Start All** button.
2. Wait for each server to fully boot (status shows as **"Running"**).
3. Verify RCON connectivity — the manager's RCON status indicators should show **connected**.

4

Enable Cross-Chat

1. Go to **Cluster Manager**.
2. Find your cluster card.
3. Click the  **Chat** button (it has a **BETA** badge).
4. The button turns **violet** when active.
5. You'll see a toast notification: *"Cross-Chat enabled"*.

That's it. Cross-chat is now active.

5

Verify It's Working

1. Join **Server A** in-game.
2. Have another player (or yourself on a second client) join **Server B**.
3. Type a message in **global chat** on Server A.
4. On Server B, you should see: `[Server A Name] PlayerName: your message`.



What Players See In-Game

When cross-chat is active, relayed messages appear in global chat with the source server name prefixed:

```
[The Island] DinoTamer99: Anyone trading Rex eggs?  
[Scorched Earth] SandWalker: I have some! Meet at the obelisk
```

[Aberration] CaveExplorer: Nice, I'll transfer over

This lets players know which map the message originated from.



Disabling Cross-Chat

1. Go to **Cluster Manager**.
 2. Click the  **Chat** button again on the cluster.
 3. The button returns to its **grey/inactive** state.
 4. Toast notification: *"Cross-Chat disabled"*.
The relay stops **immediately**. No server restart required.
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Troubleshooting

PROBLEM	CAUSE	FIX
Messages not relaying	RCON not enabled	Enable RCON in server settings & restart
Messages not relaying	Admin password is empty	Set an admin password on every server
Only some servers receive	RCON port conflict	Assign unique RCON ports to each server
No messages detected	Server not generating logs	Check that <code>ShooterGame.log</code> exists
Stops after server restart	State not auto-restored	Re-enable cross-chat after restarting servers
Button doesn't turn violet	Database/cluster issue	Toggle off and on. Check manager logs

LOG FILE PATH

The service monitors: `<ServerInstallPath>/ShooterGame/Saved/Logs/ShooterGame.log`

If your servers use a non-standard installation layout, ensure this path exists and is actively written to by the game server.

Technical Reference

Architecture Summary

- **Log watching** is non-invasive with zero FPS impact.
- **RCON broadcasting** sends a `ServerChat` command to each peer server.
- **Thread safety** uses `Arc<AtomicBool>` for safe shutdown.
- **Multi-cluster** — multiple clusters can run simultaneously without cross-talk.
- **Persistence** — enabled/disabled state stored in `cluster_settings` DB table.
- **Poll interval** — Log watcher checks for new lines in real-time (event-driven, not polling).

COMPONENT	FILE
Cross-Chat Service (backend)	<code>src-tauri/src/services/cross_chat.rs</code>
Tauri Commands	<code>src-tauri/src/commands/cross_chat.rs</code>
Cluster Toggle (DB persistence)	<code>src-tauri/src/commands/cluster.rs</code>
Frontend Toggle UI	<code>src/pages/ClusterManager.tsx</code>
Backend Documentation	<code>documentation/backend/CrossChat.md</code>